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# Feature Learning in Supervised AI Models

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## Abstract

In this research paper we will learn and investigate how various supervised learning models learn features. We will cover four path ways: Induction Biases, Domain Adaptation, Domain Generalization, and the Open Set Problem. The source code can be found on GitHub (<https://github.com/ZainFarhan-Work/ATML>).

## 1 Introduction

First, we will see the influence of Inductive Biases and Feature Representations, our goal here is to see the influence of model architecture. By comparing the metrics of four well known and standard models (ResNet-50, ViT-B, CLIP). We will first perform a baseline metric on all of them, and then perform 5 different experiments. Colour Bias, Grey Scale, Shape vs Texture, and Translation.

Then we will see Domain Adaptation and Domain Generalization, where we study DAN with MMD, DANN - Adversarial Alignment, CDAN, SAM, and DAN-DG. For the open set problem we will see GCSC and PROSER with both MLS and Placeholder

## 2 Experimental Setup/Methods

### 2.1 Inductive Biases and Feature Representations

**Dataset:** STL-10

For the Shape vs Texture we used AdaIN implementation, which follows Huang & Belongie (2017). the encoder/decoder architecture and the pretrained weights come from the public reimplementation naoto0804/pytorch-AdaIN, weights mirrored at BhupatiNadar/adain-style-transfer-models on HuggingFace.

### 2.2 Unsupervised Domain Adaptation

Data: PACS. SOurce - Photo, Art Painting, Cartoon. Target: Sketch Optimisation: AdamW, lr  $10^{-4}$ , wd  $10^{-4}$

Source Only: Cross-entropy on the pooled source batch. (Also ERM for Domain Generalisation)  
DAN:  $\mathcal{L} = \mathcal{L}_{cls} + \lambda \text{MMD}^2$  on 512-d features and kernel = sum of 3 RBFs with bandwidths  $\{0.5, 1, 2\} \times \text{median pairwise squared distance of the combined batch (bandwidth detached)}$ ,  $\lambda = 1$

### 2.3 Domain Generalisation

ERM is shared with Task 2. No sketch access.

DAN-DG:  $\mathcal{L} = \mathcal{L}_{cls} + \frac{\lambda_{DG}}{3} \sum_{e < e'} \text{MMD}^2(F(X_e), F(X_{e'}))$

Sharpness Proxy:  $\Delta_{\text{sharp}} = \mathcal{L}(\theta + \epsilon) - \mathcal{L}(\theta)$ ,  $\epsilon = 0.05 g / \|g\|_2$

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Table 1: Overall Comprehensive Comparison

| System         | Clean |       |       | Grayscale |          |       | Hue +40° |          |       | Patch shuffle |          |       |
|----------------|-------|-------|-------|-----------|----------|-------|----------|----------|-------|---------------|----------|-------|
|                | Top-1 | F1    | Conf. | Top-1     | $\Delta$ | Cons. | Top-1    | $\Delta$ | Cons. | Top-1         | $\Delta$ | Cons. |
| ResNet-50      | 96.4  | 96.39 | 0.91  | 94.6      | -1.8     | 95.8  | 94.4     | -2.0     | 95.2  | 89.0          | -7.4     | 90.4  |
| ViT-B/16       | 97.2  | 97.21 | 0.97  | 95.4      | -1.8     | 98.0  | 95.8     | -1.4     | 98.2  | 90.4          | -6.8     | 92.6  |
| CLIP + head    | 96.6  | 96.60 | 0.27  | 93.8      | -2.8     | 93.0  | 94.8     | -1.8     | 95.8  | 84.4          | -12.2    | 84.6  |
| CLIP zero-shot | 97.0  | 97.01 | 0.94  | 93.0      | -4.0     | 94.0  | 95.2     | -1.8     | 95.6  | 84.4          | -12.6    | 84.2  |

## 2.4 Open-Set Recognition

Data: CIFAR-10 train 50,000  $\rightarrow$  stratified 90/10: 45,000 train / 5,000 val, testing was done on both CIFAR 10 and CIFAR 100. Model: CIFAR ResNet-18: 3 x 3 stride-1 stem, no max-pool, random initialisation, 32 x 32 input, 512-d feature, Linear(512,10).

Scores:  $MSP = 1 - \max_k p_k$ .  $MLS = -\max_k z_k$ .  $Energy = -\log \sum_k e^{z_k}$ . Mahalanobis =  $\min_c (f - \mu_c)^\top \Sigma^{-1} (f - \mu_c)$

## 3 Results

### 3.1 Inductive Biases and Feature Representations

We first take some basic measurements as a baseline, we freeze the three classifier heads and measure top-1 accuracy, macro-F1, and mean maximum confidence (Table 12). We also do this for zero shot CLIP, we measure confidence from the softmax over scaled class similarities.

All the models perform well as a baseline, this maybe a limitation of the STL-10 dataset, as its distinct classes would be easier to classify. It should also be noted that the CLIP Max Mean is low because it uses Cosine similarity in training, and the L-2 norm forces the magnitude to be 1. So when attached with a linear head and no compensation, the value comes off as low. It does not mean it is lower in confidence than other models, just that rows can't be compared the magnitude lower, not because it is less confident than others

#### 3.1.1 Color Bias

We did two experiments with colour, the first we changed the images to greyscale measured the same metrics and got the following results (Table 13).

We can see that making the image greyscale results in a slight decrease in accuracy throughout. A more interesting observation is that there was no significant difference between Convolution and Vision Transformers, though consistency is slightly higher for the Vision Transformer. CLIP had the most sever impact. Both of the readings are lower than either ResNet or ViT-B, and further zero-shot is the lowest.

The second experiment was with a Hue Shift of 40 deg. A limitation of this is that a hue shift will only have a good effect on well saturated images. On less saturated images the change is very minimal, making the result less significant.

Again we see a minimal change in accuracy (Table 14), this time they are much closer to each other. No claims can be made as the difference is small enough that it can be considered noise. It is also of note that unlike greyscale, CLIP is much more in line with other models.

#### 3.1.2 Shape vs. Texture

For this comparison alpha strength of 1 was required, as lower values did not transfer texture well enough. We get the following results:

Out of a total of 400 candidate images only 286 were of good enough quality, which were further reduced to 220 to make each class balanced. As expected, the Vision Transformers have a considerable shape bias, while the convolutional model does not. Though one point of note is that ResNet still managed to score a 50.94 on the Shape Bias. This is considerably higher than the original Geirhos et al. (2019) paper. The reason for this can be mostly likely presumed to be the biggest limitation of this experiment.

Table 2: Shape vs Texture: Convolutional models are not shape biased.

| System                      | Balanced ( $n = 220$ ) |           |           |            |          | All accepted ( $n = 286$ ) |          |
|-----------------------------|------------------------|-----------|-----------|------------|----------|----------------------------|----------|
|                             | $N_{sh}$               | $N_{tex}$ | $N_{oth}$ | Shape bias | Coverage | Shape bias                 | Coverage |
| ResNet-50 + linear head     | 54                     | 52        | 114       | 50.94      | 48.18    | 57.78                      | 47.20    |
| ViT-B/16 + linear head      | 99                     | 14        | 107       | 87.61      | 51.36    | 89.94                      | 55.59    |
| CLIP ViT-B-32 + linear head | 116                    | 12        | 92        | 90.62      | 58.18    | 91.72                      | 59.09    |
| CLIP ViT-B-32 zero-shot     | 117                    | 7         | 96        | 94.35      | 56.36    | 95.81                      | 58.39    |

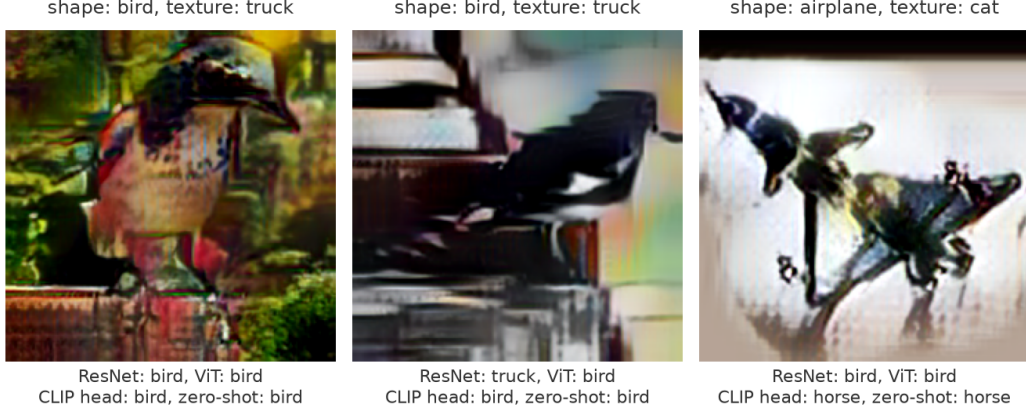


Figure 1: Example Images

71 That is that the images were filtered out manually. This introduced significant human bias in the  
72 process making the data unreliable.

73 Another notable observation is that for ViT and ResNet both, there most quantitative category is  
74 neither shape nor texture, and CLIP barely has more shape classified images. Making the coverage  
75 very low, and limiting our ability to draw further conclusions from this data. This can again be  
76 contributed to the low quality of the data. The AdaIN implementation resulted in images with quite  
77 high distortion with our particular dataset, hampering the ability of the models to classify.

78 Together it is clear that colour reliance is very low, and cue conflict shows us that shap and texture  
79 reliance is high by comparison.. However the low coverage makes further analyses and stronger  
80 claims difficult.

### 81 3.1.3 Patch Structure + Translation

82 We divided the images into 4x4 patches, and shuffled the patches. We feed the shuffled images back  
83 in the model to measure the metrics (Table 15).

84 When we shuffle the images we destroy the global structure of the image, i.e. the shape. For Vision  
85 Transformers this is quite bad as they use the self attention mechanism and look at a global picture.  
86 Understandably, they suffer the most in accuracy. Convolutional networks however only look at the  
87 local area, making them much more resilient to this, further texture is preserved more, as each patch  
88 can act like a texture patch and we saw in table2, they are much more biased towards texture.

89 The ViT-B and ResNet model are close in accuracy.

90 One final thing of note is that CLIP performs noticeably worse than ViT, even though they share the  
91 visual encoder (as a Vision Transformer), the fact that both zero-shot and linear head land at the same  
92 number implies that its not the involvement of the text encoder in the prediction process. This suggest  
93 that contrastive learning (with the text encoder) during pre training or some other stage had some  
94 factor that made them more dependent factor.<sup>1</sup>

95 The data for Translation can be seen in Figure3, from that it is clear that the spatial order is much  
96 more vital to the model than simply moving the image.

<sup>1</sup>We also saw more shape bias for them in table2, but it was not enough of a difference to warrant any conclusion

Table 3: Cosine similarities - Expected Results

| Intervention             | ResNet-50 |        | ViT-B/16 |        | CLIP ViT-B-32 |        |
|--------------------------|-----------|--------|----------|--------|---------------|--------|
|                          | $I_T$     | Norm.  | $I_T$    | Norm.  | $I_T$         | Norm.  |
| Grayscale                | 0.789     | 0.719  | 0.737    | 0.711  | 0.900         | 0.743  |
| Translation (32 px)      | 0.934     | 0.912  | 0.943    | 0.938  | 0.970         | 0.923  |
| Patch shuffle            | 0.718     | 0.626  | 0.649    | 0.615  | 0.728         | 0.299  |
| Cue conflict vs. content | 0.250     | 0.004  | 0.223    | 0.146  | 0.651         | 0.101  |
| Cue conflict vs. style   | 0.216     | -0.041 | 0.072    | -0.020 | 0.580         | -0.083 |
| Random-pair floor        | 0.247     | 0.000  | 0.090    | 0.000  | 0.612         | 0.000  |

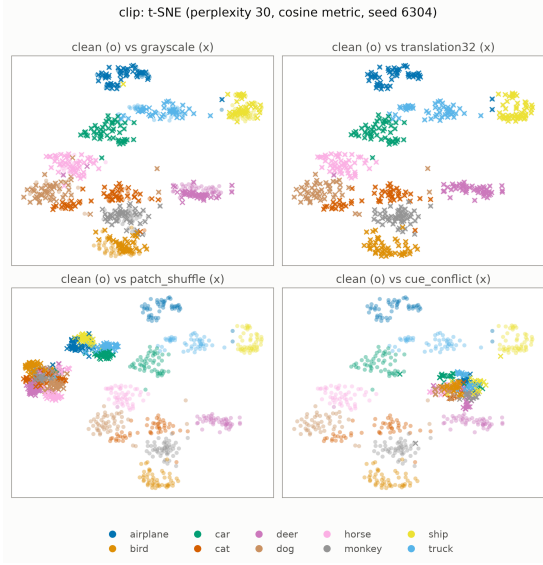


Figure 2: CLIP t-SNE: Representations overlap except in Cue Conflict and Patch Shuffle

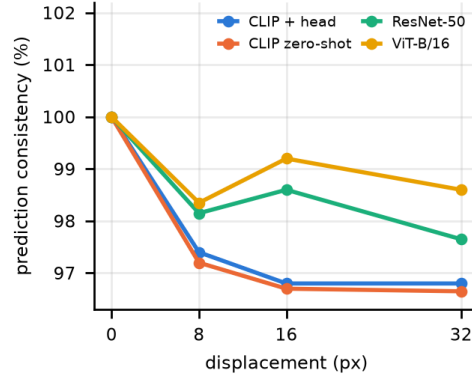


Figure 3: Translation.

### 97 3.1.4 Representation Analysis

98 We have discussed most experiments now, but those mainly measured when the predictions changed  
99 and hence also changed the accuracy. But how the model sees and interprets the feature can change  
100 even if the prediction didn't. We measure the Cosine Similarity and t-SNE maps where relevant.

101 In terms of grayscale and translation we can see the Norm is rather high. This is consistent with our  
102 earlier hypothesis that models don't look at colour for classification.

103 In Patch shuffling the values a bit lower (around 0.6), which is to be expected as the global structure  
104 is destroyed. What is more interesting is that CLIP and ViT show a large difference despite them  
105 both having a Vision Transformer. One plausible explanation is that since CLIP's tokens are twice as  
106 large as ViT, they are more dependent on global structure, and the representation drifts more. This  
107 becomes more apparent in the following diagram.

108 The rest of the t-SNE diagrams (Figures 6-7 in the appendix B.4) agree with these results.

109 Overall, a representation changes doesn't always mean a prediction change. For grayscale they all  
110 have roughly the same Norm, but there is a noticeable difference between model performance, with  
111 ViT being the most stable. We've seen that though model architecture is not the full story as the two  
112 CLIP which are close still showed significant variance in results, it is still largely important, as ViT  
113 and CLIP were very shape biased, while CNN was not.

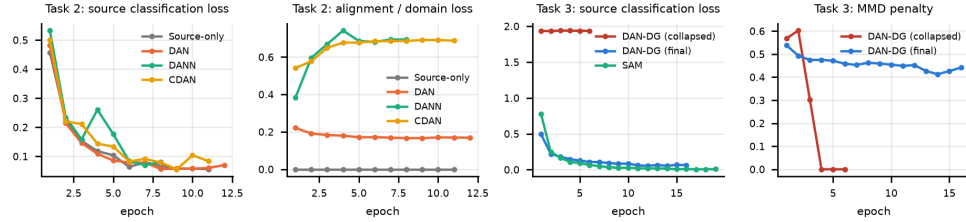


Figure 4: Training curves

## 3.2 Unsupervised Domain Adaptation

### 3.2.1 Source Only ERM

We first train both the model and the classifier here on the source domains using cross entropy, which results in these measurements (Table 16).

As we can see in source only statistics, the accuracy on our source domains is very good, but when ran on a unknown domain (for us that is sketch) we notice a massive accuracy drop. compared with the mean it is a -33 % drop, this is the main figure we try to lower in these experiments.

when we see the by class accuracy on the target domain (Table 17), dog performs worst while elephant performs best. Dog is the largest class in the sketch data, we found dog's 333 most common errors go to elephant, 167 to giraffe. It is probable it is not confusing animals and objects at least, it is confused within animal classes.

### 3.2.2 DAN - MMD Alignment

If we observe the general by class results for MMD, we see a slight but still meaningful improvement over the sketch accuracy (Table 18). Though the sketch F1 - macro decreases by a bit.

We see all classes improve in accuracy, except for giraffe and elephant (Table 19).

### 3.2.3 DANN - Adversarial Alignment

We now use adversarial learning to see how well it performs. And get the following results (Table 20).

Interestingly not only is the performance on the sketch pictures worse than source, but the the predictions on source only is also worse. At first this might make it look like DANN didn't work, but after further investigation we find this (Table 21).

The Loss was indeed decreasing, and the Domain Accuracy was going to 0.5., what this means is that even if we fool the discriminator. that does not necessarily mean an increase in performance.

### 3.2.4 CDAN - Class-Conditional Adversarial Alignment

To improve the scores from before we use the values of the prediction as well, we do a cross product with the latent and feed that to the discriminator.

The performance on the source only metric stays the same, but the accuracy on sketch images received a significant improvement (Table 22). Meaning that even soft information about the final prediction was very helpful to the discriminator. Improving its performance which improved the backbone to more aggressively compete with it. However, it was not lower domain separability that did this, as domain separability essentially remained the same.

### 3.2.5 Common Evaluation and Alignment Diagnostic

We would assume that bringing the domains closer and making them indistinguishable would be key, as the overlapping elements between the target domain and source domains would come into play. However aside from MMD which tries to directly to do this with the Loss function, a freshly trained logistical regressor can still separate the classes. And our best performance from CDAN barely shows any change, implying it might not be the most key factor.

Table 4: Diagnostic - Separability might not be the deciding factor

| Method      | Source val. acc. (%) |       |         | Mean source (%) |       | Sketch (%) |       | $\Delta$ Acc. | Sep. (%) |
|-------------|----------------------|-------|---------|-----------------|-------|------------|-------|---------------|----------|
|             | Photo                | Art   | Cartoon | Acc.            | F1    | Acc.       | F1    |               |          |
| Source-only | 97.30                | 91.71 | 95.74   | 94.91           | 94.85 | 61.59      | 63.09 | —             | 100.00   |
| DAN         | 95.80                | 93.41 | 95.95   | 95.05           | 94.93 | 64.44      | 60.55 | +2.85         | 81.87    |
| DANN        | 95.80                | 92.93 | 93.60   | 94.11           | 94.03 | 55.97      | 58.23 | -5.62         | 99.31    |
| CDAN        | 97.30                | 91.46 | 95.52   | 94.76           | 94.66 | 74.65      | 73.85 | +13.06        | 99.86    |

Table 5: Per Class Comparison: Four classes share a trend

| Class    | Sketch accuracy (%) |        |        |        |        |       | $\Delta$ vs Source-only / ERM |        |        |        |        |
|----------|---------------------|--------|--------|--------|--------|-------|-------------------------------|--------|--------|--------|--------|
|          | Source-only         | DAN    | DANN   | CDAN   | DAN-DG | SAM   | DAN                           | DANN   | CDAN   | DAN-DG | SAM    |
| Dog      | 21.11               | 47.93  | 42.10  | 64.25  | 45.60  | 36.14 | +26.82                        | +20.99 | +43.14 | +24.49 | +15.03 |
| Elephant | 94.73               | 75.54  | 35.00  | 80.54  | 90.14  | 94.73 | -19.19                        | -59.73 | -14.19 | -4.59  | +0.00  |
| Giraffe  | 77.95               | 61.62  | 51.53  | 62.95  | 49.27  | 71.85 | -16.33                        | -26.42 | -15.00 | -28.68 | -6.10  |
| Guitar   | 70.07               | 74.18  | 98.36  | 89.31  | 92.11  | 96.71 | +4.11                         | +28.29 | +19.24 | +22.04 | +26.64 |
| Horse    | 46.32               | 57.60  | 48.04  | 71.69  | 86.40  | 74.14 | +11.28                        | +1.72  | +25.37 | +40.08 | +27.82 |
| House    | 83.75               | 100.00 | 100.00 | 100.00 | 92.50  | 96.25 | +16.25                        | +16.25 | +16.25 | +8.75  | +12.50 |
| Person   | 61.25               | 86.25  | 98.12  | 99.38  | 50.00  | 75.62 | +25.00                        | +36.87 | +38.13 | -11.25 | +14.37 |

Further data shows us that four classes show interesting behavior (Table 23). Dog (+20.99 to +43.14) and Person (+25.00 to +38.13) improve under every method, while Elephant and Giraffe degrade under every method. It almost seems like the model is trading the improvement in two for the others, no one improved all four.

As alignment rises, the data converges and separability is reduced while sketch accuracy is improved a little. moderate alignment is helpful but too much destroys class information. we can use Macro-F1 without target labels as a signal.

### 3.2.6 Controlled Design Study

A higher  $\lambda$  means a higher penalty for difference between the source domain feature distribution, so increasing it should mean the model more aggressively tries to match the feature distributions. And we see exactly that for the first three rows, however  $\lambda = 10$  completely collapses the model. F1 Mean on source alone is enough to disqualify 10 as optimal even without consulting target domain labels.

We had Source-only reach 94.91% mean source accuracy against 61.59% on Sketch, a drop of 33.3 points. We saw that lower separability does not mean better prediction on target domain. And we saw class conditioning massively help in target domain accuracy.

## 3.3 Domain Generalization

We reuse the same ERM Baseline as last time, referred to in table 16 to be efficient with our compute resource.

### 3.3.1 DAN-DG - Pairwise Source-Domain Alignment

In our initial configuration, which matched task with DAN for the sake of comparison, the model collapses (Table 24).

We speculate that this happens due to the model too aggressively trying to align the source domains (Table 26). So to remedy this we try to ramp up the penalty term ( $\lambda$ ) from zero to one as the model trains instead of just starting from one. The slow increase should allow us the model to stabilise while still encouraging maximum alignment due to the increasing  $\lambda$ . Which brings us the following data (Table 25).

Table 6: Control study - too high a  $\lambda$  collapses the model

| $\lambda_{\text{MMD}}$ | Mean source F1 (%) | Sketch acc. (%) | Domain separability (%) |
|------------------------|--------------------|-----------------|-------------------------|
| 0 (source-only)        | 94.85              | 61.59           | 100.00                  |
| 0.1                    | 94.73              | 62.31           | 92.17                   |
| 1                      | 94.93              | 64.44           | 81.87                   |
| 10                     | 5.07               | 4.07            | 60.03                   |

Table 7: Full Comparison - source

| Method | Source val. acc. (%) |       |         | Mean source (%) |       | Worst (%) |       | Sketch (%) |       | $\Delta$ Acc. | Sep. (%) | $\Delta_{\text{sharp}}$ |
|--------|----------------------|-------|---------|-----------------|-------|-----------|-------|------------|-------|---------------|----------|-------------------------|
|        | Photo                | Art   | Cartoon | Acc.            | F1    | Acc.      | F1    | Acc.       | F1    |               |          |                         |
| ERM    | 97.30                | 91.71 | 95.74   | 94.91           | 94.85 | 91.71     | 91.69 | 61.59      | 63.09 | —             | 91.21    | 0.2095                  |
| DAN-DG | 96.40                | 92.93 | 94.88   | 94.74           | 94.45 | 92.93     | 92.56 | 71.49      | 73.20 | +9.90         | 70.05    | 0.2400                  |
| SAM    | 97.00                | 93.41 | 96.80   | 95.74           | 95.73 | 93.41     | 93.65 | 74.12      | 74.98 | +12.53        | 89.01    | 0.1116                  |

Table 8:  $\rho$  Values – Collapses on 0.1

| $\rho$  | Source F1 (%) |       | Sketch (%) |          | Sep. (%) | $\Delta_{\text{sharp}}$ |
|---------|---------------|-------|------------|----------|----------|-------------------------|
|         | Mean          | Worst | Acc.       | $\Delta$ |          |                         |
| 0 (ERM) | 94.85         | 91.69 | 61.59      | —        | 91.21    | 0.2095                  |
| 0.01    | 94.59         | 91.48 | 63.58      | +1.99    | 90.38    | 0.1702                  |
| 0.05    | 95.73         | 93.65 | 74.12      | +12.53   | 89.01    | 0.1116                  |
| 0.1     | 12.75         | 10.68 | 4.07       | −57.52   | 67.86    | 0.0091                  |

176 However this also collapsed, was visible from source-side metrics alone with source F1 5.07, feature  
 177 std 0, MMD 0 by epoch 4. So as a last resort we increase the Bandwidth Gradient to 1 in the RBF  
 178 Kernel, this is perhaps the most direct way of telling the MMD Loss to not bring the domains too  
 179 close, this finally results in a significant improvement to the sketch accuracy, however we can no  
 180 longer compare these results to our earlier experiments.

181 DAN-DG did make source domains less distinguishable but that was not the main driver for improve-  
 182 ment in sketch accuracy. The model completely collapsed with  $\lambda = 1$ .

### 183 3.3.2 SAM - Parameter-Space Stability

184 We now try to land on a flatter space of the loss, which should in theory be where the neighboring  
 185 space all occupy similar features.

186 This worked much better and cleaner than our DAN-DG we see a obvious improvement on our target  
 187 domain (Table 27).

### 188 3.3.3 Common Evaluation and Diagnosis

189 We compare all our numbers against each other, with one interesting trend that was also in Domain  
 190 Adaptation, which that accuracies on source domains are not at all predictive of sketch accuracy,  
 191 difference between source being minimal with SAM and ERM only, but sketch accuracy is massively  
 192 improved by SAM.

193 We see that DAN does the most work for Source Domain Separability which is what MMD targets,  
 194 while SAM does not really touch for domain separability but does reduce sharpness the most as we  
 195 want it to. SAM performs very well compared to DAN.

196 We now have also seen that mean source and and worst source are not good indicators or predictors  
 197 of performance on the target domain (sketch). SAM lowers the sharpness the most, but that only  
 198 partly tells us how well the model will perform on sketch, as we will see in the Controlled Design  
 199 Study as well. It is also hard to tell whether target aware DAN or target free DAN is better. The  
 200 best target-aware result (CDAN, 74.65) is essentially tied with the best target-free one (SAM, 74.12).  
 201 DAN and DAN-DG have difference that makes the scenario less comparable. DAN-DG tries to  
 202 bring sources together, while DAN brings source and domain together. The final DAN-DG also  
 203 lets gradient flow through the kernel bandwidth, while DAN keeps it detached, so the kernels aren't  
 204 identical.

### 205 3.3.4 Controlled Design Study

206 We varied the  $\rho$  values for SAM to find how far the trend extends

207 with high  $\rho$  values the bound extends so far that it starts to go to other regions as well. Sharp still  
 208 increases as the shallowed becomes a subset, but performance is completely destroyed which  
 209 slowest that it can't be used as an accurate independent measure of performance for SAM.

Table 9: Unknowness Score Types - Mahalanobis performed worse while Energy was Best

| Score       | AUROC (%) |       |       | At the 95% validation threshold (%) |           |          |                |
|-------------|-----------|-------|-------|-------------------------------------|-----------|----------|----------------|
|             | Near      | Far   | All   | Test accept.                        | Near rej. | Far rej. | FPR near / far |
| MSP         | 80.98     | 89.66 | 85.32 | 94.34                               | 28.25     | 45.38    | 71.75 / 54.62  |
| MLS         | 79.42     | 89.84 | 84.63 | 94.77                               | 30.62     | 54.62    | 69.38 / 45.38  |
| Energy      | 79.43     | 89.92 | 84.67 | 94.84                               | 31.38     | 54.88    | 68.62 / 45.12  |
| Mahalanobis | 79.00     | 91.22 | 85.11 | 94.62                               | 27.25     | 50.12    | 72.75 / 49.88  |

Table 10: Different Models - GCSC performs the best, both PROSER are worse than Vanilla

| Model   | Score       | CSA   | AUROC (%) |       |       | At the 95% threshold (%) |           |          |
|---------|-------------|-------|-----------|-------|-------|--------------------------|-----------|----------|
|         |             |       | Near      | Far   | All   | Test accept.             | Near rej. | Far rej. |
| Vanilla | MLS         | 94.26 | 79.42     | 89.84 | 84.63 | 94.77                    | 30.62     | 54.62    |
| GCSC    | MLS         | 95.38 | 81.84     | 92.13 | 86.99 | 94.35                    | 37.62     | 64.00    |
| PROSER  | MLS         | 93.63 | 77.61     | 87.57 | 82.59 | 94.58                    | 29.38     | 44.50    |
| PROSER  | Placeholder | 93.63 | 74.36     | 84.82 | 79.59 | 94.66                    | 27.25     | 41.37    |

Before this experiment we expected higher values of  $\rho$  to look more broadly and perhaps find shallower areas. and the sharpness value did fall, but the collapse alongside it was unexpected.

### 3.4 Open-Set Recognition (OSR)

We first train a very traditional ResNet classifier, with ten classes to get a Vanilla score. The results are this (Table 28).

We froze the Vanilla Model, and measured different unknowness scores on it (Table 29).

#### 3.4.1 GCSC - Strong Closed-Set Classifier + MLS

Training our model extremely well on a closed set should make it so that it'll be able to differentiate unknowns regardless. We achieve this by training on random positive augmentations.

For CIFAR 10 we see that both validation and test accuracy increase slightly (Table 30), but when we see the effect on unknown classifications, we see that they also improve across the board (Table 31).

#### 3.4.2 PROSER - Classifier and Data Placeholders

We make a new class by taking latents of two classes and doing a inner join to get their middle, this creates a new unknow class.

On the CIFAR-10 data set we see similar levels of performance (Table 32), though we get there earlier, the same trend continues for Unknow classifications (Table 33).

#### 3.4.3 Common Evaluation and Failure Analysis

All Unknowness score types had similar performance, in the Near rej stats, which would be the hardest, Mahalanobis performed the worst while Energy performed the best.

GCSC is the best performer, with significant improvement over Vanilla, both PROSER variants perform worse than Vanilla, the PROSER MLS can be chalked by to Non Gain from the Vanilla. The -1.25 gap could be noise. but PROSER Placeholder shows amore apprent worse score. This is more apprent in Far rejections, where both perform considerably worse than Vanilla. We choose EPOCH 14 as that is the CIFAR 10 validation was highest, but at that early stage it is possible that the PROSER simply didn'T traing enough to make significant progress, and the extra unknow class hampered the model's ability (as the logits would now be spread on one aditional class), instead of helping it.

Detecting Far imges seems to be much easier throughout, most likely because they look so different, so it's easier for the model to tell them apart. In table11 we can see that where as the worst Near are all wrongly classied at above 80%, the far are all less than 60%.

We saw it is much easier to accept near vs far, 555 of 800 near unknowns are accepted (69%), against 363 of 800 far (45%). Pickup truck, bus, and wolf got accepted most for near, while wardrobe,

Table 11: Failure Rates - Far are much easier than near

| Group | Unknown class $\rightarrow$ predicted class | MLS score | $\tau$ | Accepted in class |
|-------|---------------------------------------------|-----------|--------|-------------------|
| Near  | pickup_truck $\rightarrow$ automobile       | -11.54    | -5.72  | 91/100            |
| Near  | bus $\rightarrow$ truck                     | -12.36    | -5.72  | 82/100            |
| Near  | wolf $\rightarrow$ dog                      | -11.37    | -5.72  | 81/100            |
| Far   | wardrobe $\rightarrow$ cat                  | -11.53    | -5.72  | 58/100            |
| Far   | mushroom $\rightarrow$ cat                  | -12.73    | -5.72  | 55/100            |
| Far   | sunflower $\rightarrow$ bird                | -10.99    | -5.72  | 52/100            |

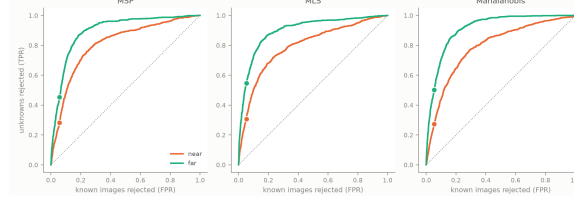


Figure 5: ROC curve for Detector - Far always performs better

242 mushrooms, and sunflower for far. pickup  $\rightarrow$  automobile, bus  $\rightarrow$  truck, tractor  $\rightarrow$  truck, wolf  $\rightarrow$   
 243 dog were plausible failures while Wardrobe  $\rightarrow$  cat, mushroom  $\rightarrow$  cat and sunflower  $\rightarrow$  bird were  
 244 surprising. MSP is normalised confidence and discards logit magnitude. MLS and Energy keep  
 245 magnitude. Mahalanobis uses feature-space distance, so it is the most independent signal. MLS and  
 246 Energy were close to each other, and MSP were close to both, with Mahalanobis being furthest. We  
 247 saw very good improvements from GCSC, while little difference from PROSER.

## 248 4 Conclusions

249 We saw that in terms of inductive bias, CNNs have strong texture bias while Visual Transformers  
 250 have a shape bias. Neither had strong colour or translation Bias. For Domain Adaptation we see DAN  
 251 with MMD beat DANN with adversarial alignment while CDAN did better than both. With Domain  
 252 Adaptation we saw SAM perform much better than DAN-DG as well as being much more stable.  
 253 Lastly, with our Open Set problem we saw GCSC significantly out perform, while both PROSER  
 254 variants were unable to beat even vanilla

## 255 5 References

## Appendix

### A More Setup and Methods

#### A.1 Inductive Biases and Feature Representations

##### Models:

- torchvision ResNet-50 with ResNet50\_Weights.IMAGENET1K\_V2;
- torchvision ViT-B/16 with ViT\_B\_16\_Weights.IMAGENET1K\_V1;
- OpenCLIP ViT-B-32 with pretrained='openai'.

##### Configuration:

- Seed: 6304
- Optimizer: AdamW
- Learning Rate:  $10^{-3}$
- Weight Decay:  $10^{-4}$

#### A.2 Table Notes

**Table 1.** Top-1, macro-F1 (F1),  $\Delta$  Top-1 and consistency (Cons.) in %; Conf. is mean max confidence.  $\Delta$  and Cons. are relative to each system’s own clean predictions. CLIP is OpenCLIP ViT-B-32; the other systems use linear heads on frozen features.

**Table 4.** Sep. = domain separability: held-out accuracy of a logistic-regression probe ( $C = 1$ , balanced, 70/30 split, seed 6304) trained to tell source from target from the frozen features. Chance is 50%. Target labels were used only at this stage.

**Table 9.** Test accept. is the share of CIFAR-10 test images accepted at the threshold.

**Table 10.** CSA is closed-set accuracy on the ten known-class logits, so both PROSER rows share it. Test accept. is the share of CIFAR-10 test images accepted at the threshold.

**Table 11.** Vanilla, MLS; accept when the score is  $\leq \tau$ . Each row is the most confidently accepted image of that unknown class; the last column gives how many of its 100 test images were accepted.

**Table 7.** ERM is Task 2’s source-only checkpoint, reused unchanged. SAM uses the specified  $\rho = 0.05$ . DAN-DG uses  $\lambda_{DG} = 1$  with gradient flowing through the median-heuristic bandwidth; with the bandwidth detached the run collapsed (Table 26).

**Table 7.** Separability: three-way logistic-regression probe over Photo / Art Painting / Cartoon on frozen features,  $C = 1$ , 70/30 split, seed 6304; chance is 33.3%.  $\Delta_{\text{sharp}}$ : cross-entropy increase after one normalised ascent step of radius 0.05 on a fixed batch of 32 images per source domain, model in eval mode.

### B Task 1: Additional Results

#### B.1 Clean Baseline

Table 12: Clean Baseline - Shows all models performing very well on the dataset.

| System                      | Top-1 (%) | Macro-F1 (%) | Mean max conf. |
|-----------------------------|-----------|--------------|----------------|
| ResNet-50 + linear head     | 96.4      | 96.39        | 0.91           |
| ViT-B/16 + linear head      | 97.2      | 97.21        | 0.97           |
| CLIP ViT-B-32 + linear head | 96.6      | 96.60        | 0.27           |
| CLIP ViT-B-32 zero-shot     | 97.0      | 97.01        | 0.94           |

Table 13: Greyscale: Minimal change in Top-1 Accuracy

| System                      | Top-1 | Macro-F1 | $\Delta$ Top-1 | Consistency | Conf. |
|-----------------------------|-------|----------|----------------|-------------|-------|
| ResNet-50 + linear head     | 94.6  | 94.61    | −1.8           | 95.8        | 0.91  |
| ViT-B/16 + linear head      | 95.4  | 95.41    | −1.8           | 98.0        | 0.96  |
| CLIP ViT-B-32 + linear head | 93.8  | 93.78    | −2.8           | 93.0        | 0.25  |
| CLIP ViT-B-32 zero-shot     | 93.0  | 92.97    | −4.0           | 94.0        | 0.93  |

Table 14: Hue Rotation of 40: Minimal change in accuracy

| System                      | Top-1 | Macro-F1 | $\Delta$ Top-1 | Consistency | Conf. |
|-----------------------------|-------|----------|----------------|-------------|-------|
| ResNet-50 + linear head     | 94.4  | 94.42    | −2.0           | 95.2        | 0.90  |
| ViT-B/16 + linear head      | 95.8  | 95.79    | −1.4           | 98.2        | 0.96  |
| CLIP ViT-B-32 + linear head | 94.8  | 94.78    | −1.8           | 95.8        | 0.26  |
| CLIP ViT-B-32 zero-shot     | 95.2  | 95.18    | −1.8           | 95.6        | 0.93  |

Table 15: Patch Shuffling: Transformers suffer the most

| System                      | Top-1 (%) | Macro-F1 (%) | $\Delta$ Top-1 | Consistency (%) | Mean max conf. |
|-----------------------------|-----------|--------------|----------------|-----------------|----------------|
| ResNet-50 + linear head     | 89.0      | 89.06        | −7.4           | 90.4            | 0.87           |
| ViT-B/16 + linear head      | 90.4      | 90.52        | −6.8           | 92.6            | 0.89           |
| CLIP ViT-B-32 + linear head | 84.4      | 84.18        | −12.2          | 84.6            | 0.20           |
| CLIP ViT-B-32 zero-shot     | 84.4      | 84.20        | −12.6          | 84.2            | 0.82           |

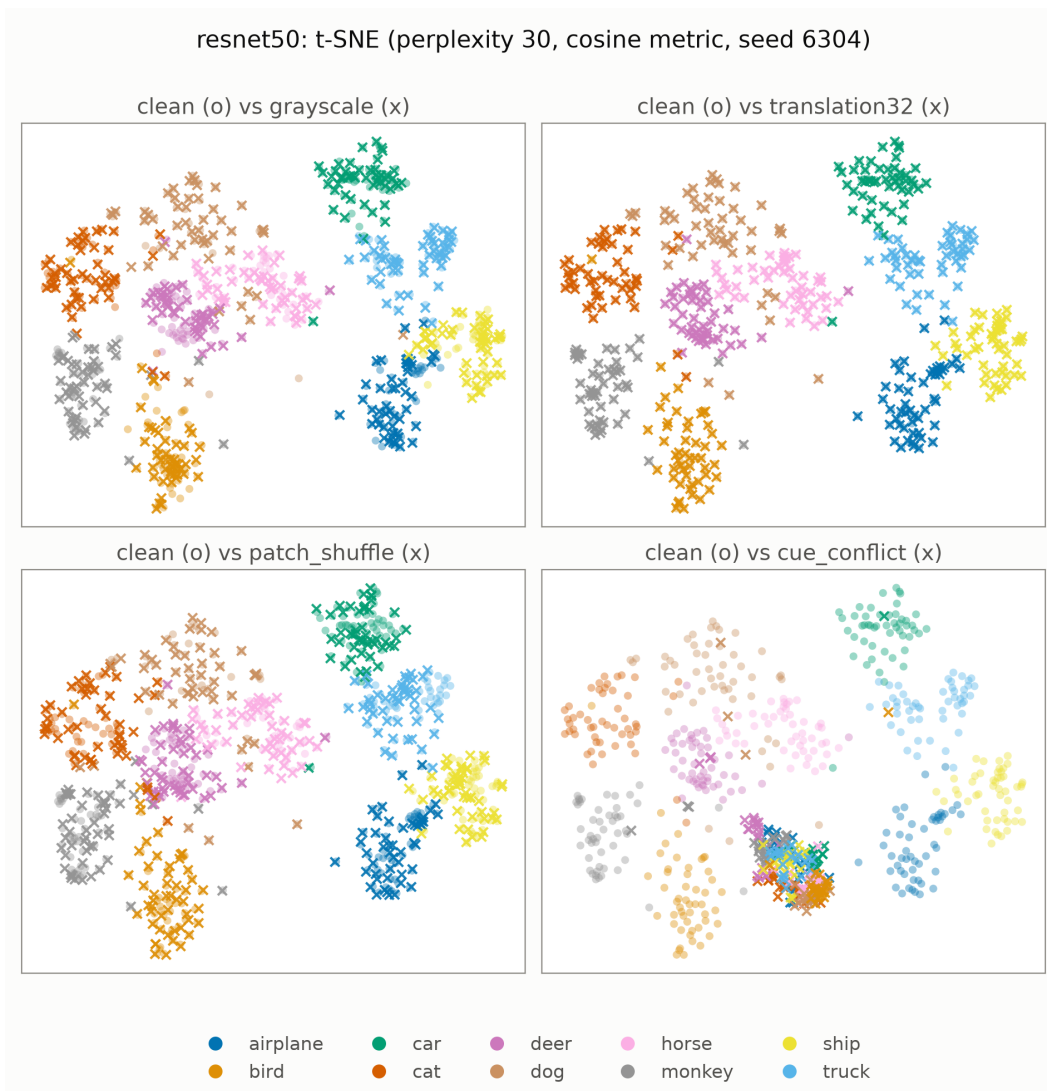


Figure 6: ResNet t-SNE: Representations overlap except in Cue Conflict

## 291 B.4 Representation Analysis

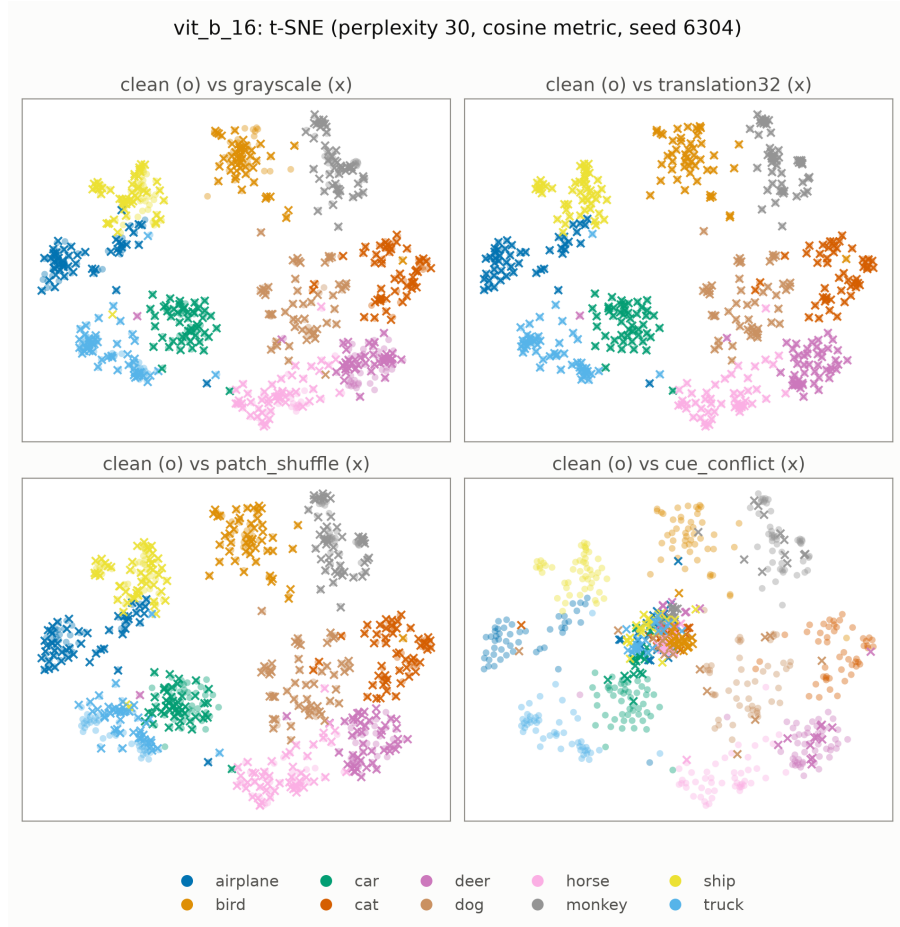


Figure 7: Vit-B t-SNE: Representations overlap except in Cue Conflict

## 292 C Task 2: Per-Method Results

### 293 C.1 Source Only ERM

Table 16: Source only ERM - the main domain gap identified

| Domain                     | Accuracy (%) | Macro-F1 (%) |
|----------------------------|--------------|--------------|
| Photo (source, val)        | 97.30        | 96.85        |
| Art Painting (source, val) | 91.71        | 91.69        |
| Cartoon (source, val)      | 95.74        | 96.00        |
| Mean source                | 94.91        | 94.85        |
| Worst source               | 91.71        | 91.69        |
| Sketch (target)            | 61.59        | 63.09        |

Table 17: Target Domain by class - high amounts of variation.

| Class    | Sketch accuracy (%) |
|----------|---------------------|
| Dog      | 21.11               |
| Elephant | 94.73               |
| Giraffe  | 77.95               |
| Guitar   | 70.07               |
| Horse    | 46.32               |
| House    | 83.75               |
| Person   | 61.25               |

## 294 C.2 DAN - MMD Alignment

Table 18: DA with MMD - Slight improvement over Source Only

|                          | Source-only | DAN ( $\lambda_{\text{MMD}} = 1$ ) |
|--------------------------|-------------|------------------------------------|
| Photo val acc (%)        | 97.30       | 95.80                              |
| Art Painting val acc (%) | 91.71       | 93.41                              |
| Cartoon val acc (%)      | 95.74       | 95.95                              |
| Mean source acc (%)      | 94.91       | 95.05                              |
| Mean source macro-F1 (%) | 94.85       | 94.93                              |
| Sketch acc (%)           | 61.59       | 64.44                              |
| Sketch macro-F1 (%)      | 63.09       | 60.55                              |
| $\Delta$ Sketch acc      | —           | +2.85                              |
| Domain separability (%)  | 100.00      | 81.87                              |

Table 19: MMD by Class - Good by class improvement

| Class    | Source-only (%) | DAN (%) | $\Delta$ |
|----------|-----------------|---------|----------|
| Dog      | 21.11           | 47.93   | +26.82   |
| Elephant | 94.73           | 75.54   | −19.19   |
| Giraffe  | 77.95           | 61.62   | −16.33   |
| Guitar   | 70.07           | 74.18   | +4.11    |
| Horse    | 46.32           | 57.60   | +11.28   |
| House    | 83.75           | 100.00  | +16.25   |
| Person   | 61.25           | 86.25   | +25.00   |

## 295 C.3 DANN - Adversarial Alignment

Table 20: DANN - The performances worsened

|                          | Source-only | DANN  |
|--------------------------|-------------|-------|
| Photo val acc (%)        | 97.30       | 95.80 |
| Art Painting val acc (%) | 91.71       | 92.93 |
| Cartoon val acc (%)      | 95.74       | 93.60 |
| Mean source acc (%)      | 94.91       | 94.11 |
| Mean source macro-F1 (%) | 94.85       | 94.03 |
| Sketch acc (%)           | 61.59       | 55.97 |
| Sketch macro-F1 (%)      | 63.09       | 58.23 |
| $\Delta$ Sketch acc      | —           | −5.62 |
| Domain separability (%)  | 100.00      | 99.31 |

Table 21: Tracking Per Epoch - Loss was decreasing while Domain accuracy was going to 0.5

| Epoch | Cls. loss | Domain loss | Domain acc. | Mean source F1 (%) |
|-------|-----------|-------------|-------------|--------------------|
| 1     | 0.533     | 0.384       | 0.817       | 91.37              |
| 2     | 0.234     | 0.597       | 0.673       | 89.19              |
| 3     | 0.159     | 0.669       | 0.597       | <b>94.03</b>       |
| 4     | 0.261     | 0.742       | 0.542       | 84.62              |
| 5     | 0.176     | 0.686       | 0.551       | 92.80              |
| 6     | 0.084     | 0.681       | 0.541       | 92.37              |
| 7     | 0.070     | 0.694       | 0.502       | 93.92              |
| 8     | 0.081     | 0.694       | 0.501       | 93.90              |

Selected epoch in bold. Chance domain accuracy is 0.5 and  $\ln 2 = 0.693$ .

#### 296 C.4 CDAN - Class-Conditional Adversarial Alignment

Table 22: CDAN: Significant Improvement from before

|                          | Source-only | CDAN   |
|--------------------------|-------------|--------|
| Photo val acc (%)        | 97.30       | 97.30  |
| Art Painting val acc (%) | 91.71       | 91.46  |
| Cartoon val acc (%)      | 95.74       | 95.52  |
| Mean source acc (%)      | 94.91       | 94.76  |
| Mean source macro-F1 (%) | 94.85       | 94.66  |
| Sketch acc (%)           | 61.59       | 74.65  |
| Sketch macro-F1 (%)      | 63.09       | 73.85  |
| $\Delta$ Sketch acc      | —           | +13.06 |
| Domain separability (%)  | 100.00      | 99.86  |

Table 23: Two most wrong - Similar pattern

| True class | Source-only               | DANN                 | CDAN                    |
|------------|---------------------------|----------------------|-------------------------|
| Dog        | elephant 333, giraffe 167 | person 397, house 18 | person 139, elephant 62 |
| Horse      | giraffe 293, elephant 111 | dog 204, person 166  | dog 134, person 33      |
| Elephant   | person 21, giraffe 13     | person 400, dog 49   | person 65, dog 58       |
| Giraffe    | elephant 99, horse 34     | person 164, dog 142  | dog 175, horse 56       |

Two most frequent wrong predictions per class, with counts.

## 297 D Task 3: Additional Results

### 298 D.1 DAN-DG - Pairwise Source-Domain Alignment

Table 24: Comparison with ERM - complete model collapsed.

|                                | ERM    | DAN-DG ( $\lambda_{DG} = 1$ ) |
|--------------------------------|--------|-------------------------------|
| Photo val macro-F1 (%)         | 96.85  | 5.86                          |
| Art Painting val macro-F1 (%)  | 91.69  | 5.14                          |
| Cartoon val macro-F1 (%)       | 96.00  | 4.21                          |
| Mean source acc (%)            | 94.91  | 21.68                         |
| Mean source macro-F1 (%)       | 94.85  | 5.07                          |
| Worst source macro-F1 (%)      | 91.69  | 4.21                          |
| Sketch acc (%)                 | 61.59  | 4.07                          |
| Sketch macro-F1 (%)            | 63.09  | 1.12                          |
| $\Delta$ Sketch acc            | —      | −57.52                        |
| Source-domain separability (%) | 91.21  | 38.74                         |
| $\Delta_{\text{sharp}}$        | 0.2095 | 0.0433                        |

Table 25:  $\lambda$  Ramp - Improvement from before but still collapsing

| Variant            | $\lambda_{DG}$ | Source (%) |          | Sketch (%) |          | Sep. (%) | $\Delta_{sharp}$ |
|--------------------|----------------|------------|----------|------------|----------|----------|------------------|
|                    |                | Mean F1    | Worst F1 | Acc.       | $\Delta$ |          |                  |
| ERM (no alignment) | 0              | 94.85      | 91.69    | 61.59      | —        | 91.21    | 0.2095           |
| As specified       | 1              | 5.07       | 4.21     | 4.07       | −57.52   | 38.74    | 0.0433           |
| Reduced weight     | 0.1            | 96.78      | 96.29    | 69.81      | +8.22    | 72.80    | 0.1394           |
| Increased weight   | 10             | 5.07       | 4.21     | 4.07       | −57.52   | 27.47    | 0.0057           |
| $\lambda$ ramp     | 0 to 1         | 94.43      | 92.41    | 62.31      | +0.72    | 76.37    | 0.3216           |
| Bandwidth gradient | 1              | 94.45      | 92.56    | 71.49      | +9.90    | 70.05    | 0.2400           |

Table 26: Step Data - Model is destroyed by step 50.

| Step | $\lambda_{DG}$ | Cls. loss | MMD   | $\ f\ $ | Feature std |
|------|----------------|-----------|-------|---------|-------------|
| 0    | 1              | 2.042     | 0.725 | 27.93   | 0.7891      |
| 50   | 1              | 1.948     | 0.286 | 3.43    | 0.0017      |
| 100  | 1              | 1.932     | 0.625 | 2.37    | 0.0003      |
| 150  | 1              | 1.925     | 0.525 | 2.10    | 0.0000      |
| 199  | 1              | 1.959     | 0.748 | 2.03    | 0.0000      |
| 199  | 0              | 0.096     | —     | 61.48   | 1.6616      |

Identical code and seed; only  $\lambda_{DG}$  differs.  $\ln 7 = 1.946$  is the cross-entropy of a uniform predictor over seven classes. Gradient clipping at 1.0 did not prevent it.

## 299 D.2 SAM - Parameter-Space Stability

Table 27: SAM - Stable performances and great improvement

|                                | ERM    | SAM ( $\rho = 0.05$ ) |
|--------------------------------|--------|-----------------------|
| Photo val macro-F1 (%)         | 96.85  | 96.41                 |
| Art Painting val macro-F1 (%)  | 91.69  | 93.65                 |
| Cartoon val macro-F1 (%)       | 96.00  | 97.13                 |
| Mean source acc (%)            | 94.91  | 95.74                 |
| Mean source macro-F1 (%)       | 94.85  | 95.73                 |
| Worst source macro-F1 (%)      | 91.69  | 93.65                 |
| Sketch acc (%)                 | 61.59  | 74.12                 |
| Sketch macro-F1 (%)            | 63.09  | 74.98                 |
| $\Delta$ Sketch acc            | —      | +12.53                |
| Source-domain separability (%) | 91.21  | 89.01                 |
| $\Delta_{sharp}$               | 0.2095 | 0.1116                |

## 300 E Task 4: Additional Results

### 301 E.1 Vanilla Closed-Set Baseline

Table 28: Vanilla Model - Intentionally made strong with 95 validation accuracy for later comparison.

| Vanilla ResNet-18                |        |
|----------------------------------|--------|
| Epochs                           | 100    |
| Selected epoch                   | 93     |
| CIFAR-10 train accuracy (%)      | 100.00 |
| CIFAR-10 validation accuracy (%) | 95.08  |
| CIFAR-10 test accuracy, CSA (%)  | 94.26  |

CIFAR-10 known classes; 45,000 / 5,000 stratified train / validation split, seed 6304. Checkpoint selected by validation accuracy only.

## 302 E.2 Post-Hoc Novelty Score

Table 29: TODO: your caption and key take-away.

| Score       | $\tau$    | Test acc. (%) | Near rej. (%) | Far rej. (%) | FPR@95TPR near / far (%) |
|-------------|-----------|---------------|---------------|--------------|--------------------------|
| MSP         | 0.1721    | 94.34         | 28.25         | 45.38        | 71.75 / 54.62            |
| MLS         | -5.7175   | 94.77         | 30.62         | 54.62        | 69.38 / 45.38            |
| Energy      | -5.9177   | 94.84         | 31.38         | 54.88        | 68.62 / 45.12            |
| Mahalanobis | 2764.0347 | 94.62         | 27.25         | 50.12        | 72.75 / 49.88            |

$\tau$  is the 95th percentile of unknownness on the CIFAR-10 validation set; accept when  $u(x) \leq \tau$ . Rejection is the share of unknowns not accepted; FPR@95TPR is the share incorrectly accepted.

## 303 E.3 GCSC - Strong Closed-Set Classifier + MLS

Table 30: GCSC - Validations acc and Test acc both increase slightly

|                                  | Vanilla | GCSC  |
|----------------------------------|---------|-------|
| Selected epoch                   | 93      | 98    |
| CIFAR-10 validation accuracy (%) | 95.08   | 95.62 |
| CIFAR-10 test accuracy, CSA (%)  | 94.26   | 95.38 |

Identical recipe, initialization, optimiser, schedule, batch size, epochs and seed; GCSC adds RandAugment(2, 9) after the crop and flip.

Table 31: GCSC - Classifications scores all improve

| Model         | AUROC near (%) | AUROC far (%) | AUROC all (%) | CSA (%) |
|---------------|----------------|---------------|---------------|---------|
| Vanilla + MLS | 79.42          | 89.84         | 84.63         | 94.26   |
| GCSC + MLS    | 81.84          | 92.13         | 86.99         | 95.38   |
| $\Delta$      | +2.42          | +2.29         | +2.36         | +1.12   |

## 304 E.4 PROSER - Classifier and Data Placeholders

Table 32: PROSER - Same Level of CIFAR-10 for less EPOCH

|                                  | Vanilla | GCSC  | PROSER |
|----------------------------------|---------|-------|--------|
| Epochs                           | 100     | 100   | 50     |
| Selected epoch                   | 93      | 98    | 14     |
| CIFAR-10 validation accuracy (%) | 95.08   | 95.62 | 94.72  |
| CIFAR-10 test accuracy, CSA (%)  | 94.26   | 95.38 | 93.63  |

PROSER is initialised from the selected Vanilla checkpoint with five dummy classifiers appended. CSA uses the ten known-class logits only.

Table 33: PROSER - Same Level of performance for less EPOCH.

| Model   | Score       | AUROC near (%) | AUROC far (%) | AUROC all (%) | CSA (%) |
|---------|-------------|----------------|---------------|---------------|---------|
| Vanilla | MLS         | 79.42          | 89.84         | 84.63         | 94.26   |
| GCSC    | MLS         | 81.84          | 92.13         | 86.99         | 95.38   |
| PROSER  | MLS         | 77.61          | 87.57         | 82.59         | 93.63   |
| PROSER  | Placeholder | 74.36          | 84.82         | 79.59         | 93.63   |